

CURIONEST CHEMISTRY

CHEMISTRY FOUNDATIONS

Lab Safety, Equipment & Measurement



GHS safety



Graduated cylinder



Electronic balance



Erlenmeyer flask

PRINT + EDITABLE

Teacher guide | visual reference pages | stations | practical | quiz | full key

Grades 9-11 • 3-5 class periods • US Letter

START HERE

Teacher Guide

A flexible 3-5 day launch unit for Grades 9-11

PURPOSE Students build safe lab habits, recognize essential equipment, and make defensible measurements before the first wet lab.

Day	Focus	Suggested evidence
1	Safety norms + Safety Detective	Correct unsafe actions using evidence
2	GHS labels + SDS navigation	Locate hazard, PPE, first-aid information
3	Equipment identification + tool choice	Match tool to task and justify
4	Measurement stations	Read meniscus; tare and report mass
5	Practical + quiz	Demonstrate procedures and explain choices

Prep checklist

- Print pages labeled Student Page or Assessment; retain Teacher Page and Teacher Key pages.
- Set out clean, empty equipment only. Do not use chemicals for the identification stations.
- Test the eyewash and safety shower according to school policy; never activate them solely as a student demonstration.
- Use local emergency procedures and the SDS for every chemical used in later labs.

STANDARDS ROLE This foundational skills unit supports NGSS science practices such as planning investigations, analyzing data, and using mathematics. It is not presented as a standalone HS-PS1 performance-expectation unit.

Suggested materials

Goggles, lab apron, beaker, Erlenmeyer flask, graduated cylinder, test tube and rack, dropper, funnel, electronic balance, wash bottle, ring stand, hot plate, and printed station cards.

SAFETY NOTE This resource supports instruction; it does not replace district policy, teacher training, equipment inspection, or chemical-specific SDS review.

STUDENT PAGE 1

Before You Touch the Tools

Student name: _____ Class: _____ Date: _____

Complete the readiness check. Your teacher may add local procedures.

Readiness area	What safe practice looks like
Eye protection	Wear chemical splash goggles whenever chemicals, heat, or glassware are used as directed.
Clothing	Closed-toe shoes; secure loose hair and clothing; wear the assigned apron.
Behavior	Read the full procedure first. No food, drink, horseplay, or unauthorized experiments.
Emergency	Know exits, gas/electric shutoffs, eyewash, shower, extinguisher, and spill-response procedures.
Chemicals	Read the label and SDS. Never return unused chemical to the stock container.
Incidents	Report every spill, breakage, exposure, injury, or near miss immediately.

Explain, do not just agree

Why must splash goggles remain over the eyes rather than on the forehead?

Why should unused chemicals never be returned to the stock bottle?

Student commitment: I will stop and ask when a procedure, label, or safety direction is unclear.

Student signature: _____ Date: _____

Lab Safety Detective

Read eight classroom snapshots. Explain the safer action for each.

LAB SAFETY DETECTIVE

Eight classroom snapshots • Identify the unsafe choice before writing the safer action.

1 EYE PROTECTION

A student begins work with splash goggles resting on the forehead.

What should happen instead?

2 FOOD & DRINK

A drink cup sits beside the lab notebook and glassware.

What should happen instead?

3 LABELING

A beaker contains blue liquid but has no chemical label.

What should happen instead?

4 HEAT

Loose paper is placed beside a lit laboratory burner.

What should happen instead?

5 BROKEN GLASS

Broken glass is left on the bench after cleanup begins.

What should happen instead?

6 HOUSEKEEPING

A backpack blocks the aisle beside the lab station.

What should happen instead?

7 CHEMICALS

An open reagent bottle is left unattended on the bench.

What should happen instead?

8 PERSONAL SAFETY

Long hair hangs loose near the work area and heat source.

What should happen instead?

#	Unsafe choice	Safer action
1		
2		
3		
4		
5		
6		
7		
8		

GHS Pictogram Reference

Symbols communicate hazard classes; they do not replace the full label or SDS.

GHS Hazard Pictograms

 Exploding bomb Explosives; some self-reactives/peroxides	 Flame Flammable and related fire hazards	 Oxidizer Oxidizing gases, liquids, or solids
 Gas cylinder Gases under pressure	 Corrosion Skin burns/eye damage; corrosive to metals	 Skull Fatal or toxic acute toxicity
 Exclamation Irritant or harmful effects	 Health hazard Cancer, organ, respiratory, aspiration hazards	 Environment Aquatic toxicity; not OSHA-mandated

REMEMBER OSHA requires eight pictograms in its Hazard Communication Standard. The environmental pictogram is shown because students may see it, but OSHA does not mandate it.

STUDENT PAGE 4

Read a Chemical Label and SDS

Use the fictional acetone label below, then identify where the SDS supplies more detail.

SAMPLE CHEMICAL LABEL

ACETONE
Signal word: **DANGER**



- Highly flammable liquid and vapor.
- Causes serious eye irritation.
- Keep away from heat, sparks, and open flame.
- Wear eye protection; use with adequate ventilation.

Supplier: Example Chemical Co. | Emergency: 555-0100

Read the SDS before use. This fictional label is for instruction only.

1. Which pictograms appear, and what hazards do they communicate?

2. List two precautions stated on the label.

3. Which SDS section would you check for first-aid measures?

4. Which SDS section describes exposure controls and personal protection?

5. Why is a pictogram alone not enough information for safe use?

SDS QUICK MAP 1 Identification | 2 Hazard(s) | 4 First aid | 7 Handling/storage | 8 Exposure controls/PPE | 10 Stability/reactivity | 13 Disposal

STUDENT PAGE 5

Equipment Reference

Study the name and function. Circle tools available in your classroom.

Lab Equipment Reference - Set A

**Beaker**

Holds/mixes liquids; approximate volume

**Erlenmeyer flask**

Swirls mixtures with less splashing

**Graduated cylinder**

Measures liquid volume accurately

**Test tube**

Runs small-scale reactions

**Buret**

Delivers precise variable volumes

**Dropper pipet**

Transfers small amounts dropwise

STUDENT PAGE 6

Equipment Reference

Study the name and function. Circle tools available in your classroom.

Lab Equipment Reference - Set B

**Funnel**

Guides pouring or supports filtration

**Electronic balance**

Measures mass; tare before use

**Hot plate**

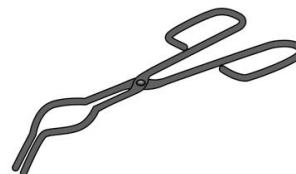
Heats without an open flame

**Wash bottle**

Dispenses distilled/deionized water

**Ring stand**

Supports clamps and apparatus

**Crucible tongs**

Handles hot crucibles

STUDENT PAGE 7

Equipment Identification Challenge

Write the correct name for each numbered tool.

Equipment Identification Challenge

**Odd-numbered tools**

1. _____

3. _____

5. _____

7. _____

9. _____

11. _____

Even-numbered tools

2. _____

4. _____

6. _____

8. _____

10. _____

12. _____

STUDENT PAGE 8

Choose the Right Tool

Name the best tool and justify why it fits the task.

Task	Best tool	Reason
Measure 24.6 mL of water		
Determine the mass of a metal sample		
Add liquid one drop at a time		
Heat a solution without an open flame		
Support a buret during titration		
Transfer a solid into a narrow-neck flask		
Rinse the inside wall of glassware		
Handle hot glassware safely		

Compare precision

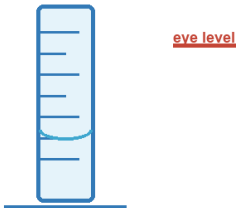
Rank these from least to most appropriate for measuring 25.0 mL: beaker, graduated cylinder, buret. Explain your ranking.

STUDENT PAGE 9

Measurement Skills

Choose the instrument first; the number of reported digits follows the instrument.

Choose the Tool - Read the Scale - Report the Unit

BEAKER	GRADUATED CYLINDER	ELECTRONIC BALANCE
Approximate volume	Measured volume	Mass after taring
		
Do not claim cylinder-level precision	Read bottom of meniscus	Container on -> TARE -> sample
Value: _____ Unit: _____	Value: _____ Unit: _____	Value: _____ Unit: _____

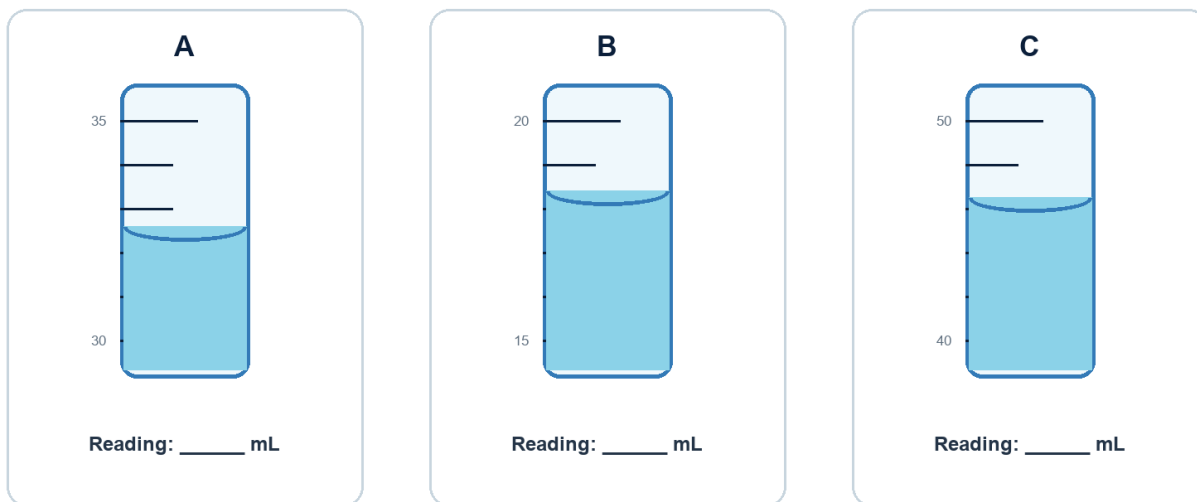
THREE HABITS Read at eye level. Record every certain digit plus one estimated digit for an analog scale. Always include a unit.

STUDENT PAGE 10

Meniscus and Precision Practice

Read the bottom of each meniscus at eye level.

Read Each Graduated Cylinder



1. Which cylinder permits the most precise reading? Explain using the scale divisions.

2. A student records Cylinder A as 33 mL. What important measurement information is lost?

3. Describe one source of parallax error and how to prevent it.

TEACHER PAGE

Lab Skills Circuit - Teacher Setup

Four low-risk stations using clean equipment and water only.

Station	Setup	Performance evidence
A: Identify	Display 8 tools	Students name each and state one safe use.
B: Volume	Cylinder + colored water	Students read volume at eye level.
C: Mass	Balance + empty container + sample	Students tare, measure, and report unit.
D: Tool choice	Scenario cards	Students select a tool and justify precision/safety.

Teacher moves

- Demonstrate one non-example before students rotate.
- Require students to say both the value and unit aloud.
- Do not place chemicals at these stations.
- Use the practical rubric on page 15 for rapid feedback.

DIFFERENTIATION Provide the reference pages at the first rotation; remove them for the second. For advanced students, add a precision comparison between graduated and volumetric glassware.

Lab Skills Circuit - Student Record

Name: _____ Group: _____

Station	Observation / measurement	Evidence or reason
A	Tool 1: _____ Tool 2: _____	Safe use: _____
B	Volume: _____ mL	Eye-level evidence: _____
C	Container mass after tare: _____ g Sample mass: _____ g	Why tare? _____
D	Scenario: _____ Chosen tool: _____	Why this tool? _____

Exit reflection

One habit that prevents an inaccurate measurement is...

One safety action I can now explain—not merely memorize—is...

ASSESSMENT

Lab Skills Practical

Teacher observation rubric; score each criterion 0-2.

Criterion	2 - Meets	1 - Developing	0 - Not yet
PPE and readiness	Correct PPE; area clear; reads task first	Minor prompt needed	Unsafe/not ready
Tool identification	Names tool and appropriate use	Name or use incomplete	Incorrect
Volume technique	Eye level; bottom meniscus; unit	One technique error	Multiple errors
Balance technique	Tares; waits for stable reading; unit	One technique error	Multiple errors
Explanation	Uses safety/precision evidence	Partial reason	No defensible reason

Score: _____ / 10 Teacher feedback: _____

REASSESSMENT Allow a second attempt after feedback. The goal is safe, repeatable performance—not a one-time score.

ASSESSMENT

Foundations Quiz - Part A

Name: _____ Class: _____

1. Which tool is best for measuring 18.6 mL?

A. Beaker B. Graduated cylinder C. Erlenmeyer flask D. Watch glass

2. What should you do first after a chemical splashes into your eyes?

A. Finish the procedure B. Use the eyewash and alert the teacher C. Wipe eyes with a towel D. Look up the formula

3. Which pictogram indicates an oxidizer?

A. Flame over circle B. Gas cylinder C. Exclamation mark D. Skull and crossbones

4. Why is a balance tared?

A. To heat the sample B. To subtract the container mass C. To change grams to mL D. To increase sample mass

5. Which SDS section contains first-aid measures?

A. Section 1 B. Section 2 C. Section 4 D. Section 10

6. Which action is safest?

A. Smell directly from a bottle B. Return excess chemical to stock C. Tie back loose hair D. Wear goggles on the forehead

Match the tool to the task

Task	Tool
7. Add liquid dropwise	_____
8. Support a buret	_____
9. Heat without an open flame	_____
10. Measure mass	_____

ASSESSMENT

Foundations Quiz - Part B

Explain your reasoning using safety or measurement evidence.

11. A student reads a graduated cylinder while looking down from above. Identify the likely error and describe the correction.

12. The label shows a flame pictogram and the word DANGER. State two precautions that should be checked before use.

13. A classmate wants to measure 25.0 mL with a beaker because it is faster. Write a claim, evidence, and reasoning response.

14. During cleanup, you find broken glass in the sink. Explain the safe response.

TEACHER KEY 1

Answer Key - Safety and GHS

Suggested responses; accept equivalent scientifically sound wording.

Item	Suggested answer
Safety Detective 1-4	1 goggles over eyes; 2 remove drink; 3 label container; 4 move paper/flammables away and control flame.
Safety Detective 5-8	5 notify teacher; use broken-glass bin with proper tools; 6 remove backpack from aisle; 7 do not leave a reagent container open or unlabeled; 8 secure loose hair.
Label Q1	Flame = flammable hazard; exclamation = irritant/harmful effects.
Label Q2	Any two: keep away from ignition; wear eye protection; use adequate ventilation.
Label Q3-4	First aid = SDS Section 4. Exposure controls/PPE = Section 8.
Label Q5	The label is a quick warning; the SDS provides detailed controls, first aid, storage, stability, and disposal information.

Readiness explanations

Goggles must cover the eyes to protect against splashes. Chemicals are not returned to stock because they may contaminate the original reagent.

TEACHER KEY 2

Answer Key - Equipment and Measurement

Use the reference pages to reteach before reassessment.

#	Equipment	#	Equipment
1	Beaker	7	Funnel
2	Erlenmeyer flask	8	Electronic balance
3	Graduated cylinder	9	Hot plate
4	Test tube	10	Wash bottle
5	Buret	11	Ring stand
6	Dropper pipet	12	Crucible tongs

Choose the Right Tool

Task	Answer	Reason
24.6 mL	Graduated cylinder	Graduations support a measured volume.
Mass	Electronic balance	Tare container; report g.
Dropwise	Dropper pipet	Controls small additions.
Heat/no flame	Hot plate	No open flame.
Support buret	Ring stand	Holds a clamp and buret upright.
Solid to narrow flask	Funnel	Guides transfer.
Rinse glassware	Wash bottle	Directed stream.
Hot glassware	Crucible tongs	Keep hands away from hot surfaces.

Precision ranking: beaker < graduated cylinder < buret for variable-volume measurement. Suitability still depends on the specific task and required uncertainty.

TEACHER KEY 3

Answer Key - Measurement, Quiz and Sources

Suggested answers and scoring notes for consistent feedback.

Meniscus practice: A = 32.6 mL; B = 18.4 mL; C = 46.5 mL. Cylinder A/B have 1 mL minor divisions, so readings should include an estimated tenths place. Cylinder C has 2 mL divisions and supports an estimated reading between marks. Parallax occurs when the eye is above or below the meniscus; move to eye level.

Quiz	Answer / scoring note
1-6	B, B, A, B, C, C
7-10	Dropper pipet; ring stand; hot plate; electronic balance
11	Viewing from above causes parallax; place eyes level with the meniscus and read the bottom.
12	Any two defensible label/SDS precautions, such as remove ignition sources, wear goggles, or provide ventilation.
13	Claim: use a graduated cylinder. Evidence: its scale supports a more precise reading than a beaker. Reasoning links smaller divisions to lower reading uncertainty.
14	Notify the teacher; do not pick up by hand; use designated tools and place glass in the broken-glass container per local procedure.

Teacher safety references

American Chemical Society. Student Laboratory Code of Conduct; Lab & Safety Equipment; Importance of the Laboratory Experience. OSHA. Hazard Communication Standard / GHS Questions and Answers.

<https://institute.acs.org/acs-center/lab-safety/education-training/high-school-labs/student-lab-code-of-conduct.html>

<https://www.acs.org/education/policies/middle-and-high-school-chemistry/classroom-and-lab-facilities/safety-equipment.html>

<https://www.osha.gov/hazcom/faq>

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